04/09/2019:

Code Analysis (Sonarqube)

Introduction:

① Sonarqube is an opensource tool suite to measure and analyse the quality of source code.

② It is written in java but able to analyse code in about 20 different programming languages.

③ Code analysis may be started manually by executing a so-called sonar runner but SonarQube full is potential especially revealed when used in combination with continous integration such as Jenkins servers.

④ The results of an analysis are shown in a fancy web frontend with 'green' and 'red lights', charts, issue lists and an ability to drill down from a project level to a single class

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Prerequisites to install Sonarqube in a windows environment:

① jdk 8

② git

③ maven

④ jenkins.

Sonarqube installation steps in windows environment:

① Google search sonarqube download and click on first link.

② Click on show all versions download dropdown

(5.6.7)

③ select sonarqube 7.8 and click on developer edition in community edition dropdown

④ Go to downloads and extract the sonarqube 7.8 to c:\programfiles.

⑤ Set the classpath in environment variables:

→ Variable name: SONAR_HOME
Variable path: <path of sonarqube 7.8>

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6. Open cmd prompt with run as admin
7. type cd / and press enter.
8. type cd c:\ProgramFiles\sonarqube-7.8\bin\windows-x86-64
9. type dir
10. type InstallNTService.bat (for first time only)
10. Type StartNTService.bat
11. Type Start Sonar.bat.
12. Open browser enter:

    → localhost:9000

14. click on login
15. Enter default username and pswd

    → username: admin
    password: admin.

Integration of sonar in 2 ways:

1. Raw way with Maven
2. with jenkins.

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Sonarqube configuration in windows env using maven:

① Navigate to c:\Users\Ravikumar\.m2

② Create a file settings.xml

③ Google search settings.xml for sonar and click on first link.

④ Copy and paste the code in settings.xml file and do some changes.

⑤ Remove the pluginsgroup

⑥ Add the following in properties group

→ sonar.host.url : http: localhost:9000

→ <sonar.login> admin</sonar.login>

→ <sonar.password> admin </sonar.passwords>

⑦ Save the file

⑧ open ravilogin2 folder

⑨ Right click → gitbash here

⑩ run the following comands.

→ mvn compile

→ mvn package

→ mvn sonar:sonar

⑪ Open sonarqube browser and refresh & check

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Sonarqube configuration in windows env (using jenkins)

① Google search sonarqube runner and click on first link

② click on download link

③ Go to downloads and extract the sonar-runner-dist-2.4 to c:\programfiles.

④ Set the classpath in environment variable

→ Variable name : SONAR_RUNNER_HOME
→ Variable value : <path of sonar-runner-dist 2.4>

⑤ In browser: localhost:8080

↓

login to jenkins &

install sonarqube plugin.

Restart jenkins.

⑥

Manage Jenkins

↓

Configure system

↓

Sonarqube servers

↓

Add sonarqube

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→ Name: mysonar (any name)

→ Server URL: http://localhost:9000

Apply & Save

(7) Manage Jenkins

↓

Global tool configuration

↓

Sonarqube scanner

↓

Add sonarqube scanner

↓

Uncheck the install automatically option (*)

↓

(8) Name: mysonarrunner

sonar-runner-home: <path of sonar runner>

→ c:\Program Files\sonar-runner-2.4

↓

Apply & Save.

(9) Create a job: freestyle

Git

↓

build

↓

clean install

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(11) Select Sonarqube scanner from add build step dropdown.

(12) In Analysis properties text area, add the following:

sonar.login = admin
sonar.password = admin
sonar.projectKey = <any name>
sonar.projectName = <same name>
sonar.projectVersion = 1.0
sonar.sources = . (current directory)

↓

Apply & Save

(13) Build now.

↓

Open sonarqube dashboard & refresh

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